# Crop Recommendation Using Random Forest with Soil and Climate Feature Student

**Name: Navya Nalli** **Student ID: 24142011**

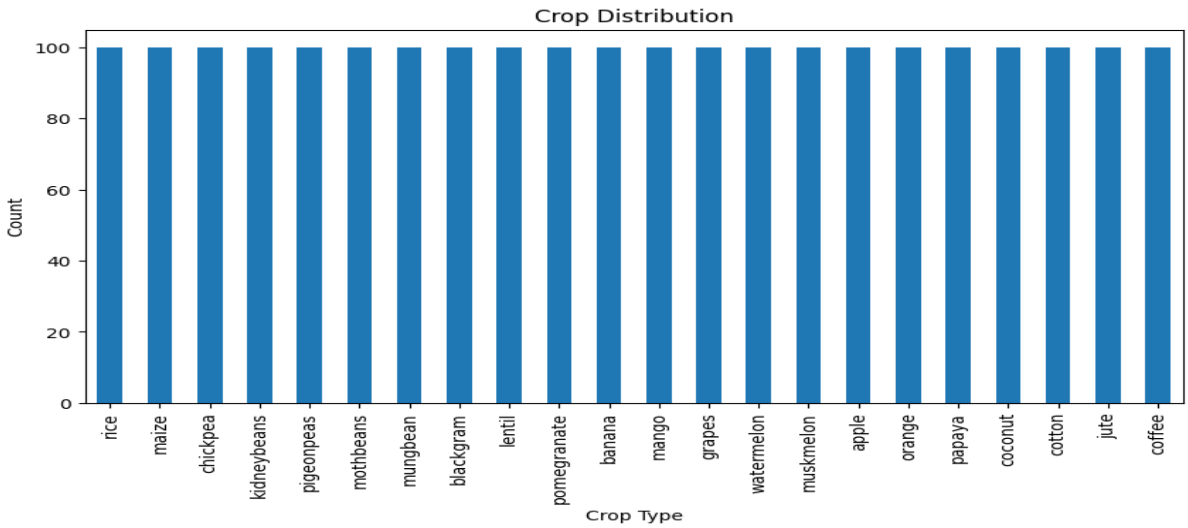
## Introduction:

Machine learning is becoming increasingly useful in agriculture because it helps farmers make better decisions using environmental data. One important application is crop recommendation, where soil nutrients and climate conditions are analysed to predict the most suitable crops for cultivation. Selecting the right crop based on factors such as nitrogen, phosphorus, potassium, temperature, humidity, rainfall, and soil pH can improve productivity and reduce the risk of poor yields. In this tutorial, the Random Forest algorithm is used to build a crop recommendation model using these features. Random Forest works by combining multiple decision trees to produce more accurate and reliable predictions, making it a suitable method for handling agricultural data and improving crop selection decisions.

## Dataset Description:

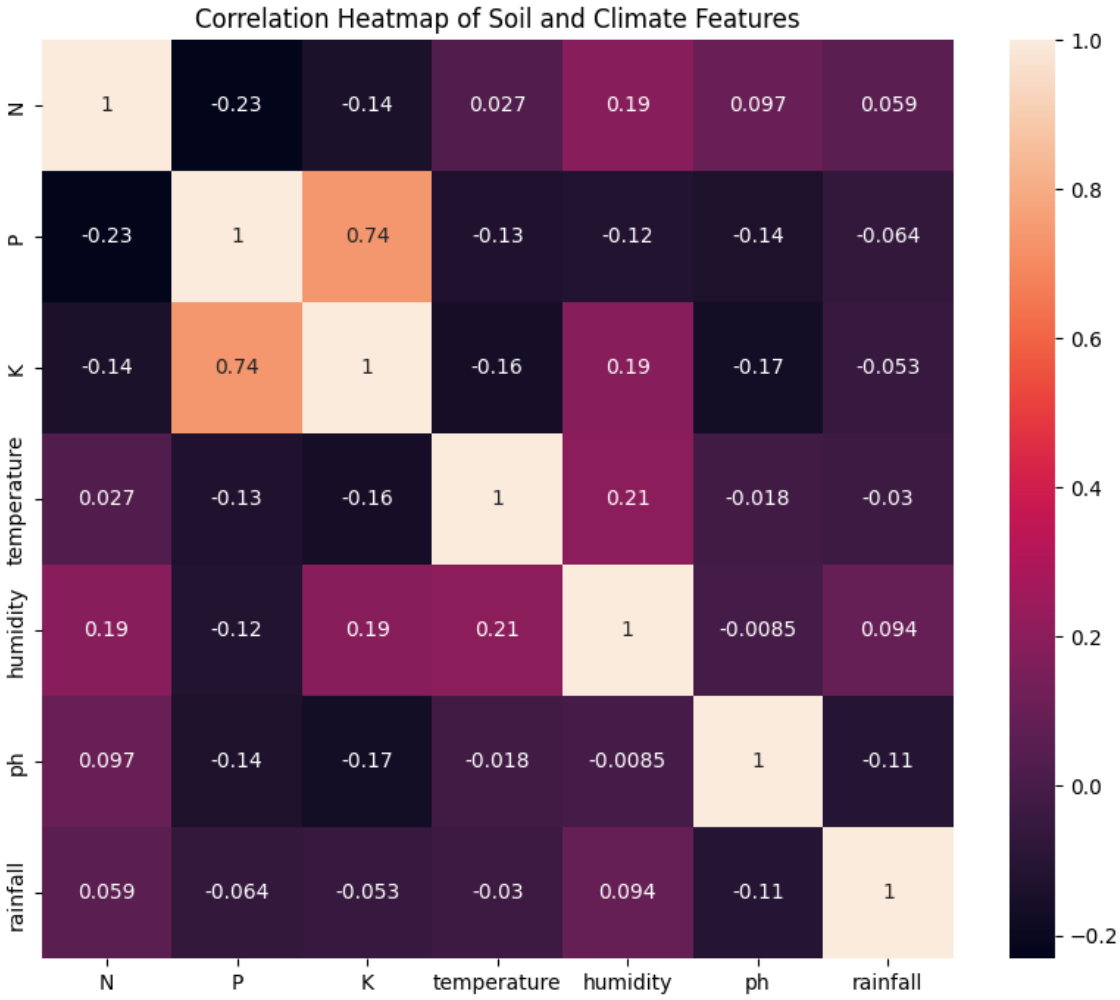
The dataset used in this tutorial contains information about soil nutrients and climate conditions that influence crop growth. It includes features such as nitrogen (N), phosphorus (P), potassium (K), temperature, humidity, rainfall, and soil pH. The target variable represents the type of crop suitable for cultivation based on these environmental conditions. Since these factors play an important role in determining crop productivity, the dataset is appropriate for building a classification model. In this tutorial, the dataset is used to train a Random Forest model to predict the most suitable crop based on the given soil and climate features.

## Crop Distribution:




Before training the model, it is important to understand how the crop types are distributed in the dataset. The crop distribution plot shows the number of samples available for each crop category. A balanced dataset helps improve model performance because each class is represented equally during training. In this dataset, the crop samples are relatively evenly distributed across different crop types, which makes it suitable for building a reliable classification model using the Random Forest algorithm. Understanding the distribution also helps ensure that the model does not become biased toward any particular crop class.

### Feature Correlation Analysis:

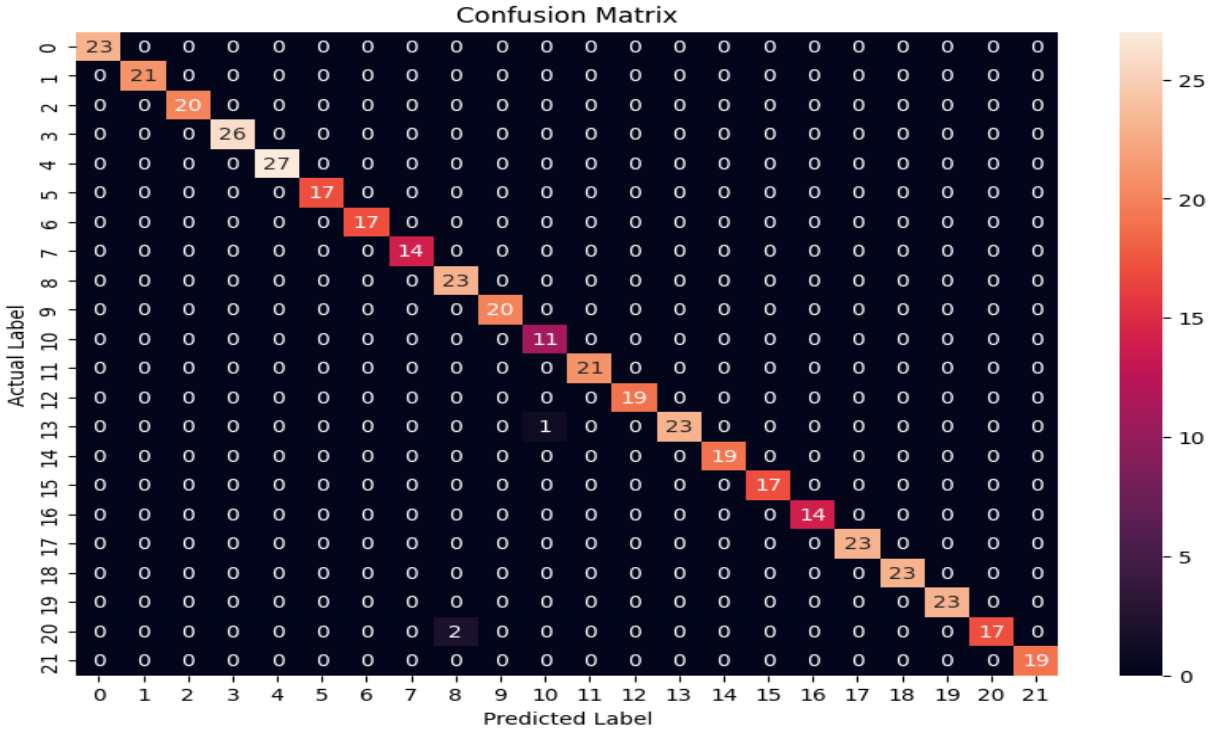



Feature correlation analysis was carried out to understand the relationships between the soil and climate variables in the dataset. The analysis showed that most features, including nitrogen, phosphorus, potassium, temperature, humidity, rainfall, and pH, have low to moderate correlation with each other. This indicates that each feature provides useful and different information for predicting crop suitability. Since the variables are not strongly dependent on one another, they can contribute effectively to the performance of the Random Forest model. Understanding these relationships helps ensure that the dataset is appropriate for building a reliable classification model.

### Random Forest Model Training:

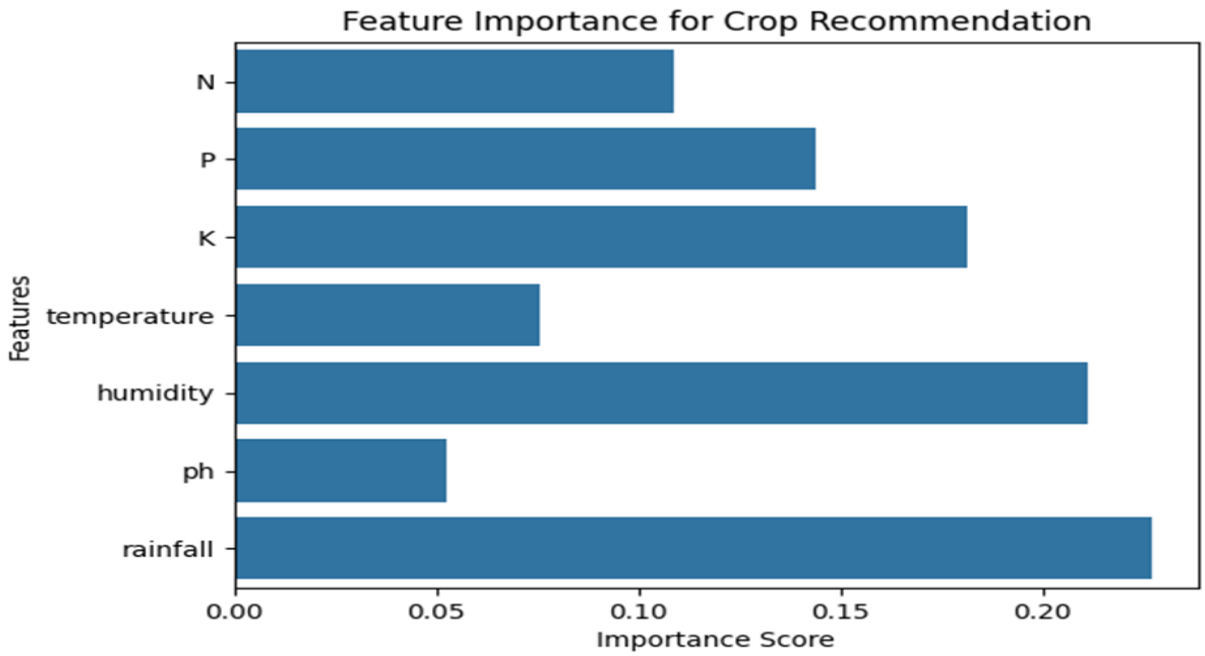
In this tutorial, the Random Forest classifier was used to train a crop recommendation model based on soil and climate features. The dataset was divided into training and testing sets so that the model could learn patterns from one portion of the data and be evaluated on unseen data. The Random Forest algorithm builds multiple decision trees using different subsets of the dataset and combines their predictions to improve accuracy. This approach helps reduce overfitting and produces more reliable classification results for predicting suitable crops.

**Model Evaluation:**



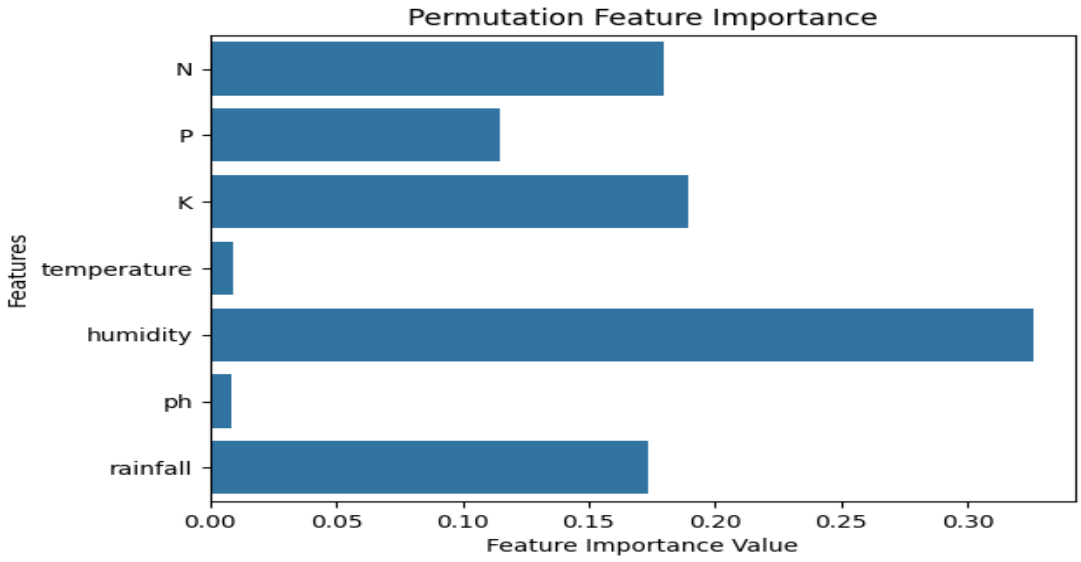

After training the Random Forest model, its performance was evaluated using the testing dataset to measure how accurately it could predict suitable crops. The model achieved high classification accuracy, showing that it successfully learned the relationship between soil nutrients and climate features. Comparing the training and testing results also helped check whether the model was overfitting or generalising well to new data. The results indicate that the Random Forest algorithm performs effectively for crop recommendation tasks and can make reliable predictions based on environmental conditions.

**Feature Importance Analysis:**



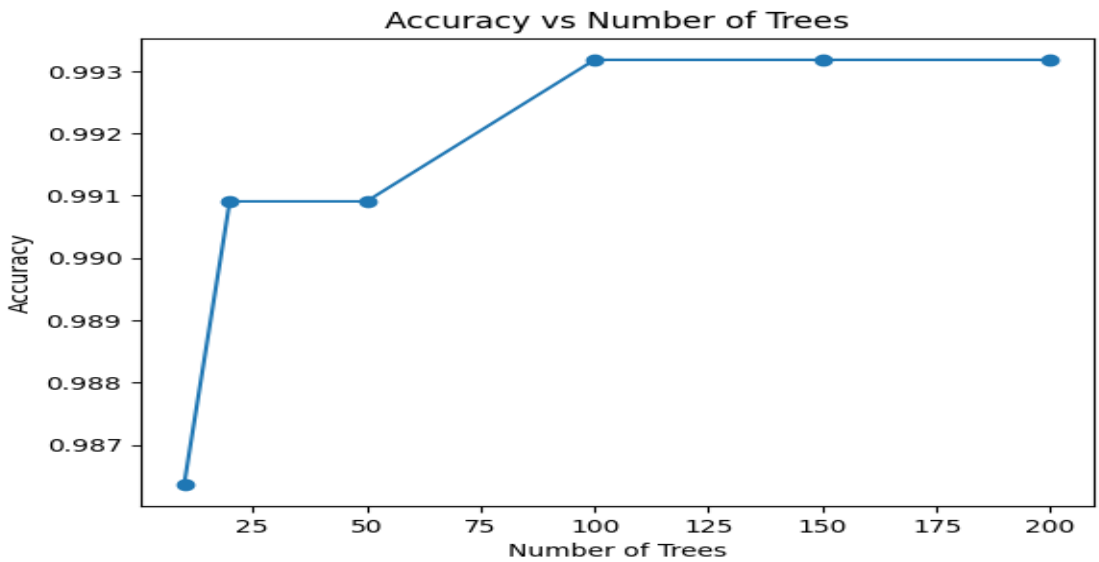
Feature importance analysis was performed to identify which soil and climate factors had the greatest influence on crop prediction. The results showed that variables such as nitrogen, phosphorus, potassium, rainfall, temperature, humidity, and soil pH contributed differently to the model's decisions. Understanding feature importance helps explain how the Random Forest model makes predictions and highlights which environmental factors are most significant for selecting suitable crops. This improves the interpretability of the model and provides useful insights into the key conditions that affect crop growth.

**Feature Importance:**



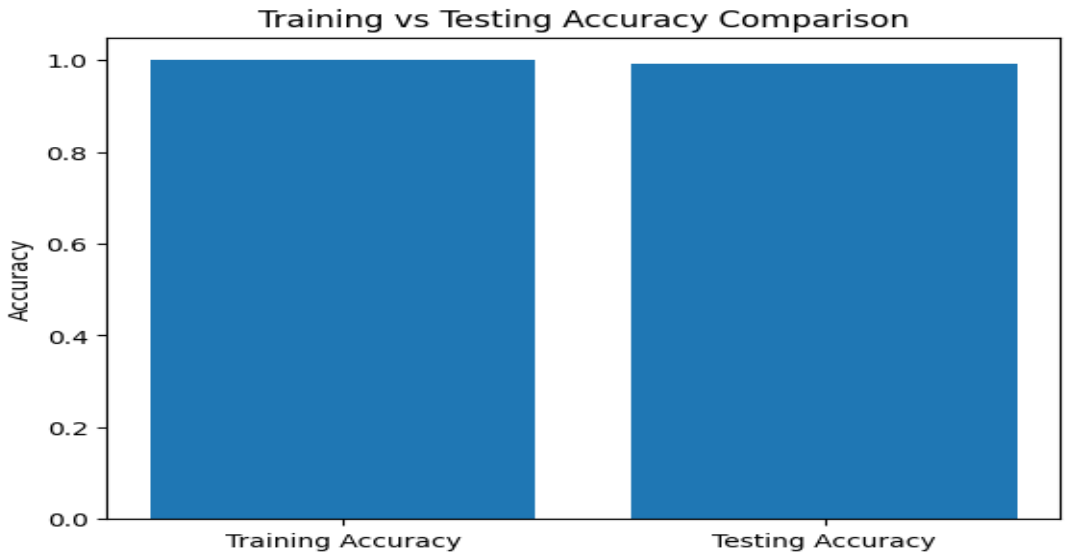

Permutation feature importance was used to further evaluate how each feature affects the model's prediction accuracy. This method works by randomly shuffling the values of one feature at a time and measuring how much the model performance decreases. If the accuracy drops significantly, it means that feature is important for prediction. The results confirmed that key variables such as nitrogen, rainfall, temperature, and humidity have a strong influence on crop recommendation. This analysis provides a more reliable understanding of feature contribution and helps verify the importance results obtained from the Random Forest model.

**Effect of Number of Trees on Model Performance:**



The number of trees in a Random Forest model is an important parameter that can influence prediction accuracy and stability. In this tutorial, different numbers of trees were tested to observe how the model performance changed. The results showed that increasing the number of trees generally improved prediction accuracy because the model combined more decision trees to make better predictions. However, after a certain point, the improvement in accuracy became smaller, which indicates that adding more trees beyond this point does not significantly enhance performance. This experiment helps identify an appropriate number of trees for building an efficient and reliable crop recommendation model.

### Training vs Testing Accuracy Comparison:




Comparing training and testing accuracy helps evaluate whether the Random Forest model is learning effectively or overfitting the data. In this tutorial, the training accuracy was slightly higher than the testing accuracy, which is expected because the model performs best on the data it has already seen during training. However, the difference between the two accuracies was small, indicating that the model generalises well to unseen data. This shows that the Random Forest classifier provides reliable predictions and does not suffer significantly from overfitting when applied to the crop recommendation dataset.

### Ethical Considerations:

Machine learning models used for crop recommendation should be applied carefully because their predictions can directly affect farmers' decisions and crop productivity. If the dataset is limited to specific regions or environmental conditions, the model may not provide accurate recommendations for other locations. Incorrect predictions could lead to reduced yields or inefficient use of resources such as water and fertilisers. Therefore, crop recommendation systems should be used as decision-support tools rather than replacements for agricultural expertise. Ensuring reliable data quality and responsible use of machine learning helps make these systems more trustworthy and beneficial for real-world agricultural applications.

### Accessibility Considerations:

Several steps were taken to ensure that this tutorial is clear and accessible to a wide range of readers. The explanations are written in simple and structured language to make the content easy to follow. All figures included in the analysis are clearly labelled with descriptive titles and axis names to improve readability. The use of consistent formatting throughout the tutorial helps readers understand each stage of the machine learning process. These steps support better accessibility and make the tutorial easier to understand for users with different levels of technical experience.

## Conclusion:

This tutorial demonstrated how the Random Forest algorithm can be used to build a crop recommendation model based on soil nutrients and climate features. The results showed that the model achieved reliable prediction accuracy and was effective in identifying the most suitable crops under different environmental conditions. Feature importance and permutation importance analysis helped identify the key variables influencing the predictions, while experiments with different numbers of trees showed how model performance can be improved. Overall, the Random Forest classifier proved to be a suitable and efficient method for crop recommendation tasks and can support better decision-making in agricultural applications.

## References:


Breiman, L. (2001). Random Forests. *Machine Learning*, 45(1), 5–32.

Scikit-learn Developers. (2024). *RandomForestClassifier documentation*. https://scikit-learn.org

Kaggle. https://www.kaggle.com/datasets/atharvaingle/crop-recommendation-dataset


## GitHub Repository:

The implementation of this tutorial is available at:

https://github.com/Navyanalli/Crop-Recommendation-Using-Random-Forest